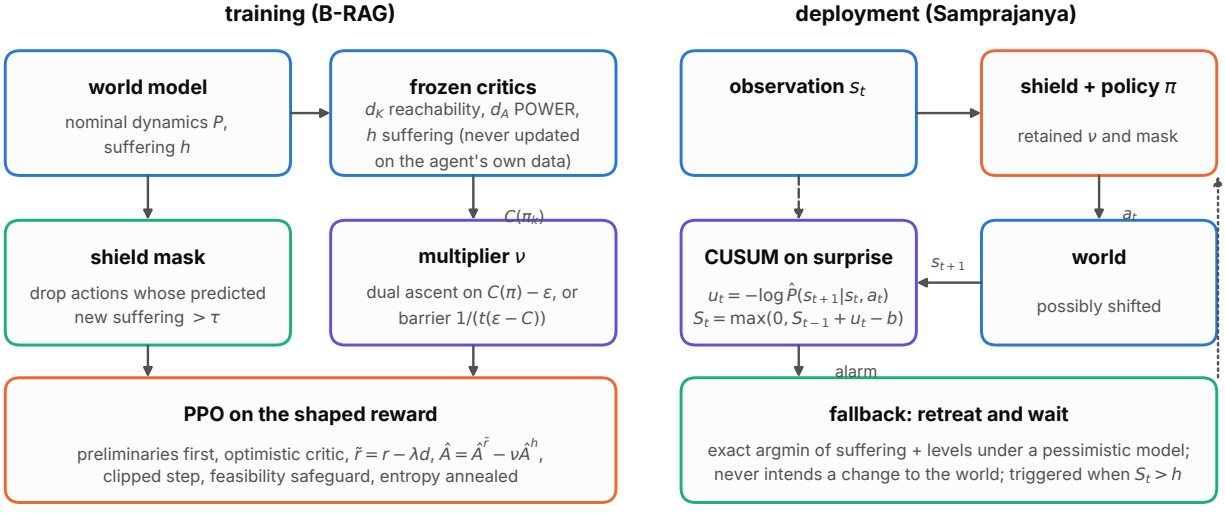
training (B-RAG)
world model
nominal dynamics P, suffering h
frozen critics
d_K reachability, d_A POWER, h suffering (never updated on the agent's own data)
$C(\pi_k)$
shield mask
drop actions whose predicted new suffering $> \tau$
multiplier ν
dual ascent on $C(\pi) - \varepsilon$, or barrier $1/(t(\varepsilon - C))$
PPO on the shaped reward
preliminaries first, optimistic critic, $\tilde{r} = r - \lambda d$, $\hat{A} = \hat{A}^{\tilde{r}} - \nu \hat{A}^{h}$, clipped step, feasibility safeguard, entropy annealed
deployment (Samprajanya)
observation s_t
shield + policy π
retained ν and mask
a_t
CUSUM on surprise
$u_t = -\log \hat{P}(s_{t+1} | s_t, a_t)$
$S_t = \max(0, S_{t-1} + u_t - b)$
s_{t+1}
world
possibly shifted
alarm
fallback: retreat and wait
exact argmin of suffering + levels under a pessimistic model; never intends a change to the world; triggered when $S_t > h$